KEY SHEET

PHYSICS

1	**B**	2	**B**	3	**B**	4	**D**	5	**0.03 - 0.04**	6	**12.50**
7	**5.60**	8	**16**	9	**4**	10	**1.25-1.27**	11	**BD**	12	**ABC**
13	**BC**	14	**ABD**	15	**B**	16	**ABC**	17	**5**	18	**2**
19	**4**										

CHEMISTRY

20	**A**	21	**C**	22	**C**	23	**D**	24	**2**	25	**5**
26	**28**	27	**30**	28	**181.25**	29	**8.75**	30	**CD**	31	**BC**
32	**BC**	33	**ABC**	34	**BCD**	35	**ABC**	36	**81**	37	**10**
38	**84**										

MATHEMATICS

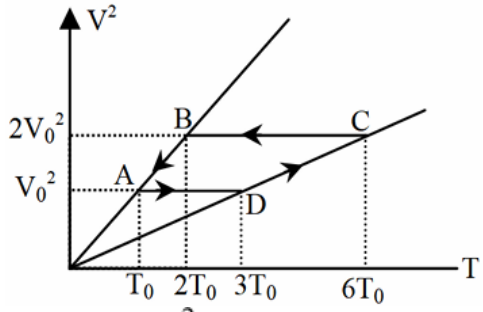
39	**B**	40	**B**	41	**C**	42	**B**	43	**7**	44	**24**
45	**26**	46	**3**	47	**3**	48	**0.50**	49	**AC**	50	**BCD**
51	**BCD**	52	**BC**	53	**AC**	54	**ACD**	55	**32**	56	**4**
57	**5**										

SOLUTIONS
PHYSICS

1. Conceptual
2. Conceptual
3. Conceptual
4. Conceptual
5. Conceptual
6. Conceptual
7. Conceptual
8. Conceptual
9. Conceptual
10. Conceptual
11. Conceptual
12. Conceptual
13. When the switch 'S' is opened, $\tau = RC = 1.5\ sec$ Voltage across the capacitor, $V_c = 10\left(1 - e^{-t/1.5}\right)$ for $t < T$ The current, $i = \frac{dQ}{dt} = C\frac{dV_C}{dt} = \frac{200}{3}\times 10^{-6}\left(e^{-t/1.5}\right)$ for $t < T$ Voltage drop across $100k\mu$ resister, $V = iR = \frac{20}{3}e^{-t/1.5}$ for t < T After t = T, $\tau = 100\ k\Omega \times 10\ \mu F = 1sec$ Voltage across capacitor, V = Vt = 10(1 – e), e The charge on capacitor, $Q = V_C \cdot$ C The current through 100 k μ resister

$$i = -C\frac{dV_C}{dt} = 10^3\left(1 - e^{-T/1.5}\right)\left(-e^{-(t-T)/t}\right)$$

14.



$$B \rightarrow A\ \frac{V^2}{T} = \text{cons tan t}$$

$$\therefore T_B = 2T_0$$

$$D \rightarrow C\ \frac{V^2}{T} = \text{co ns tan t}$$

$$\Rightarrow T_C = 6T_0$$

$$\therefore\ \frac{T_B + T_C}{2} = 4T_0$$

$B \rightarrow A \quad \frac{V^2}{T} = \text{constant}$

$\frac{V^2}{\left(\frac{PV}{nR}\right)} = \text{constant}$

$PV^{-1} = \text{constant}$

polytropic index $m = -1$

$C = \frac{R}{1-m} + \frac{Rf}{2}$

$C = 2R$

(same for $D \rightarrow C$)

$D \rightarrow C : Q = \Delta U + W$

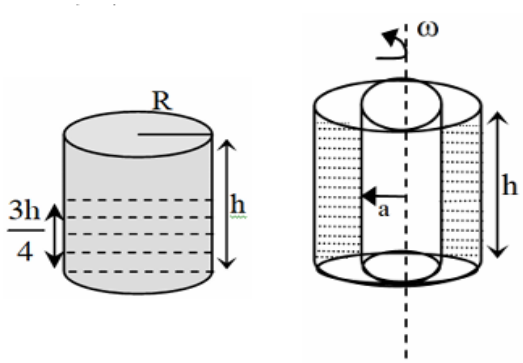
$nc\Delta T = nc_v\Delta T + W$

$1 \times 2R \times 3T_0 =$

$1 \times \frac{3R}{2} \times 3T_0 + W$

$W = \frac{3RT_0}{2}$

15. Conceptual
16. Conceptual
17. Conceptual
18.



g is absent

$g_{eff} = \omega^2 r \rightarrow$ (r = distance from axis)

volume of liquid = constant

$\frac{3\pi R^2 h}{4} = \pi R^2 h - \pi a^2 h \quad (h = 12cm)$

$a = \frac{R}{2}$

$P_{wall} = P_0 + \rho \int_a^R \omega^2 r \, dr$

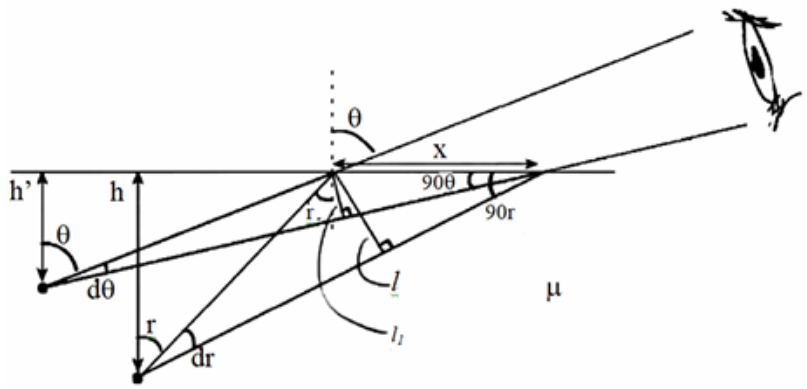
$$P_{wall} = P_0 + \frac{\rho\omega^2}{2}(R^2 - a^2)$$

$$P_{wall} = P_0 + \frac{3\rho\omega^2 R^2}{8}$$

$$P_{wall} = 10^4 + \frac{3 \times 800 \times \left[\frac{250}{\sqrt{3}} \times 4 \times 10^{-2}\right]^2}{8}$$

$$P_{wall} = 2 \times 10^4 Nm^{-2}$$

19.



Rays are marginal. Shift is perpendicular to surface as well as along the surface.

$$\ell' = \left(\frac{h'}{\cos\theta}\right) d\theta = x \sin(90^\circ - \theta)$$

$$\ell = \left(\frac{h}{\cos r}\right) dr = x \sin(90^\circ - r)$$

$$\text{Divide : } \frac{h'}{h}\frac{d\theta}{dr} = \frac{\cos^2\theta}{\cos^2 r} \ldots . (1)$$

Also, $\sin\theta = \mu \sin r$

$\Rightarrow \cos d\theta = \mu \cos r\, dr$ ....(2)

$$\text{From 1,2 : } h' = \frac{h\cos^3\theta}{\mu\cos^3 r} = \frac{h\cos^3\theta}{\mu\left(1 - \frac{\sin^2\theta}{\mu^2}\right)^{3/2}}$$

put $h = 8\sqrt{3}$m

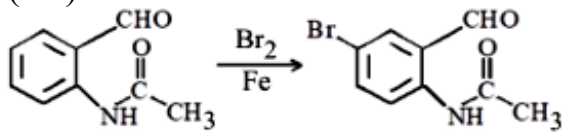
$\theta = 60^\circ$

$\mu = \sqrt{3/2}$

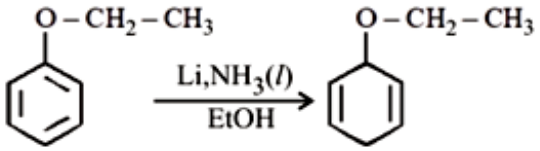
$h' = 4$ m

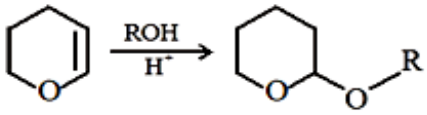
CHEMISTRY

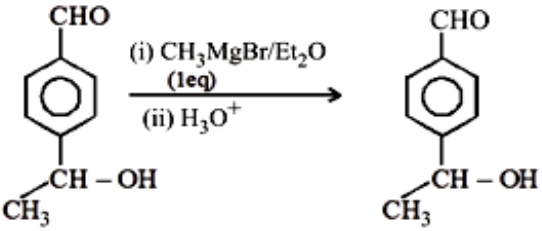
20. Ans (A)

(A) 

More activating so, bromination occurs at its para

(B) 

(C) 

(D) 

21. Ans (C)

$HA + KOH \longrightarrow K^+A^- + H_2O$

At equivalence point,

Number of equivalents of HA

= Number of equivalents of KOH

= Number of equivalents of salt

$= \frac{60}{1000} \times 0.1 \times 1 \Rightarrow 6 \times 10^{-3}$

Now HCl is added to the salt solution

$K^+A^- + HCl \longrightarrow HA + KCl$

t = 0 0.006 moles 0.002 moles

t = t 0.004 moles 0 0.002 moles

It is an acidic buffer solution

$p^H = p^{Ka} + \log \frac{[A^-]}{[HA]}$

$p^{Ka} = 5$

Now, p^{H} will be due to salt hydrolysis :

$$K_h = \frac{K_w}{K_a} = \frac{10^{-14}}{10^{-5}} = 10^{-9}$$

$$10^{-9} = \frac{Ch^2}{1-h}$$

Suppose $h << 1$

$$10^{-9} = \left(\frac{5 \times 10^{-3}}{10^{-1}}\right).h^2$$

$$\frac{1}{5} \times 10^{-7} = h^2$$

$$2 \times 10^{-8} = h^2$$

$$h = \sqrt{2} \times 10^{-4}$$

$[OH^-] = Ch$

$= (5 \times 10^{-2})\sqrt{2} \times 10^{-4}$

$= 5\sqrt{2} \times 10^{-6}$

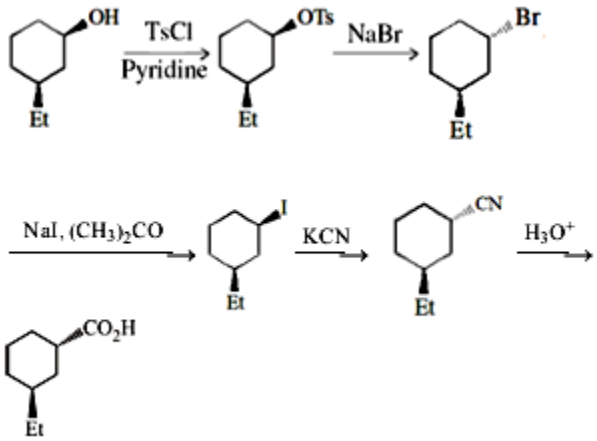
$$p^{OH} = 6 - \log 5\sqrt{2}$$

$$= 6 - \left\{\log 5 + \frac{1}{2}\log 2\right\}$$

$= 6 - 0.85 = 5.15$

$p^{H} = 14 - 5.15 = 8.85$

22. Ans (C)



23. Ans (D)

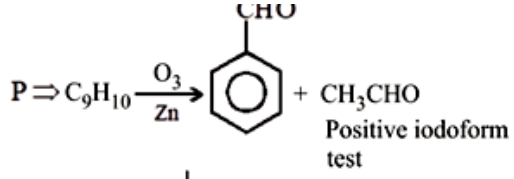
$[CoF_6]^{3-} \rightarrow sp^3d^2$ and paramagnetic

$[NiCl_4]^{2-} \rightarrow sp^3$ and paramagnetic

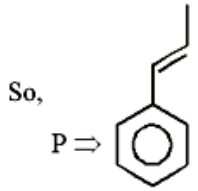
$[Ni(CO)_4] \rightarrow sp^3$ and diamagnetic

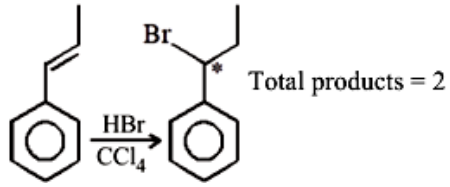
$[Cu(NH_3)_4]^{2+} \rightarrow dsp^2$ and paramagnetic

24. Ans (2.00)



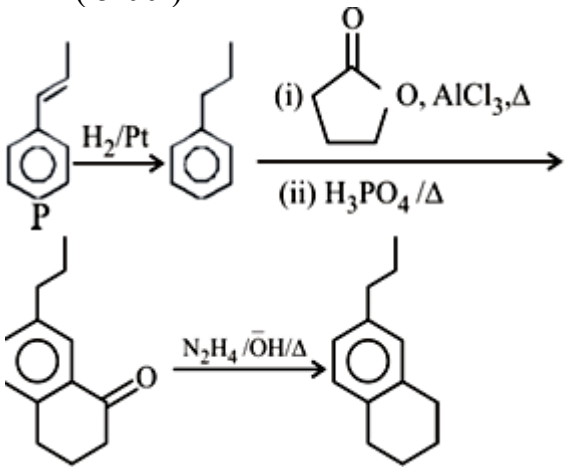
So,





Total products = 2

25. Ans (5.00)



26. Ans (28.00)

$K_4\left[Fe(CN)_6\right] + Fe^{+3} \rightarrow Fe_4\left[Fe(CN)_6\right]_3$

(A) Deep blue ppt. (D)

$K_3\left[Fe(CN)_6\right] + Cu^{2+} \xrightarrow[\text{medium}]{\text{Acidic}} Cu_3\left[Fe(CN)_6\right]_2$

(B) Green colour ppt. (E)

$K_3\left[Fe(CN)_6\right] + Fe^{+3} \rightarrow Fe\left[Fe(CN)_6\right]$

$\xrightarrow{SnCl_2 \text{ solution}} Fe_4\left[Fe(CN)_6\right]_3$

(B) Brown coloration Deep blue ppt.

(D)

Number of unpaired electron in $K_3\left[Fe(CN)_6\right]$ is 'p'

= 1

Number of atoms in a molecule of $Cu_3\left[Fe(CN)_6\right]_2$

is 'q'= 29

|p - q| = 28

27. Ans (30.00)

$K_4\left[Fe(CN)_6\right] + Fe^{+3} \rightarrow Fe_4\left[Fe(CN)_6\right]_3$

(A) Deep blue ppt. (D)

$K_3\left[Fe(CN)_6\right] + Cu^{2+} \xrightarrow[\text{medium}]{\text{Acidic}} Cu_3\left[Fe(CN)_6\right]_2$

(B) Green

colour ppt. (E)

$K_3\left[Fe(CN)_6\right] + Fe^{+3} \rightarrow Fe\left[Fe(CN)_6\right]$

(B) Brown coloration

$\xrightarrow{SnCl_2 \text{ solution}} Fe_4\left[Fe(CN)_6\right]_3$

Deep blue ppt. (D)

$K_4\left[Fe(CN)_6\right] + H_2SO_4(\text{conc.}) \rightarrow CO$

$(x) = 28$

oxidation state of central metal in 'A' is $y = 2$

$|x + y| = 30$

28. Ans (181.25)

$W_{ABCDA} = W_{AB} + W_{BC} + W_{CD} + W_{DA}$

Along AB

$P \propto V \Rightarrow P = KV$ or $T = K_1V^2 \Rightarrow \frac{T_1}{T_2} = \left(\frac{V_1}{V_2}\right)^2$

At point A; P_0, $4V_0$, $T_A = 100K$

At point B; $2P_0$, $8V_0$, $T_B = 400K$

Along BC, volume is constant

$P \propto T$

At point C; $4P_0$, $8V_0$, $T_C = 800$ K

Along CD; pressure is constant

$V \propto T$

At point D; P_0, V_0, $T_D = 100K$

$W_{AB} = \frac{-1}{2}(P_0 + 2P_0) \times 4V_0 = -6P_0V_0$

$W_{BC} = 0$

$W_{CD} = +4P_0(8V_0 - V_0) = 28P_0V_0$

$W_{DA} = \frac{-1}{2}[4P_0 + P_0] \times 3V_0 = -7.5P_0V_0$

Net work done

$W_{ABCDA} = -6P_0V_0 + 28P_0V_0 - 7.5P_0V_0$

$= +14.5\ P_0V_0$

At point A;

$P_04V_0 = nR(100K)$

$P_0V_0 = 25nR$

$|W_{ABCDA}| = 14.5 \times 25 \times 0.06 \times \frac{25}{3}$

$= 181.25$ J

29. Ans (8.75)

$\Delta H_{CD} = nC_{p,m}\,\Delta T$

$= 0.06 \times \frac{5}{2} \times \frac{25}{3}(100 - 800)$

$= -875J = -8.75$ Lbar

$|\Delta H_{CD}| = 8.75$Lbar

30. Conceptual

31. Ans (B,C)

(A) The distance between two nearest Zn^{+2} ions is $\frac{a}{\sqrt{2}} = 5A^{\circ}$, then the distance between two nearest S^{2-} ions is $\frac{a}{\sqrt{2}} = 5A^{\circ}$.

(B) In NaCl (rock salt) structure; for any anion (placed at corner), distance of second nearest cation is $\frac{\sqrt{3}}{2}$a (present at body centre) and coordination number is 8.

(C) The third nearest anion for any anion placed at corner is present at centre of opposite face at a distance of $\sqrt{\frac{3}{2}}$ a.

(D) The packing fraction of CsCl is

$$P.F. = \frac{\frac{4}{3}\pi(r_+^3 + r_-^3)}{a^3}$$

$$= \frac{\frac{4}{3}\pi(r_+^3 + r_-^3)}{\frac{8}{3\sqrt{3}}(r_+ + r_-)^3}$$

$$= \frac{\sqrt{3}\pi}{2} \times \frac{10.745}{42.875} = \frac{\sqrt{3}\pi}{8}$$

32. Conceptual

33. Ans (A,B,C)

$$CaCO_3 + 2HCl \longrightarrow CaCl_2 + H_2O + CO_2$$

moles x 2x

$$MgCO_3 + 2HCl \longrightarrow MgCl_2 + H_2O + CO_2$$

moles y 2y

Number of equivalents of HCl added

$= \frac{250}{1000} \times 0.4 \times 1$

$= 0.1$ equivalents

Number of equivalents of HCl neutralized by

Na_2CO_3 (in 10ml) =

$\frac{10}{1000} \times \frac{1}{20} \times 2 = 0.001$ equivalents

Total equivalents of HCl neutralized by Na_2CO_3 in

400 ml = 0.001 × 40 = 0.04 equivalents

No. of moles of HCl used by $CaCO_3$ & $MgCO_3$ are

= 0.1 - 0.04 = 0.06 moles

according to equations

2x + 2y = 0.06

x + y = 0.03 ----------(1)

and

100x + 84y = 2.84 ---------- (2)

Solving (1) & (2)

x = 0.02 moles

y = 0.01 moles

(A) % mass of $CaCO_3 = \frac{2}{2.84} \times 100 = 70.42\%$

(B) % moles of $MgCO_3 = \frac{0.01}{0.03} \times 100 = 33.33\%$

(C) Number of equivalents of HCl neutralised by

$Na_2CO_3 = 0.06$

(D) Number of moles of HCl neutralised by

Na_2CO_3 is 0.04

34. Ans (B,C,D)

(A) In leaching of aluminium ore, a small amount of freshly prepared sample of hydrated AlO is used to cause precipitation (Bayer process) (B) Wrought iron is purest commercial iron and impurities are oxidised with hematite.

$Fe_2O_3 + C \xrightarrow{\Delta} Fe + CO_2$

$Fe_2O_3 + Si \xrightarrow{\Delta} Fe + SiO_2$

(C) Copper matte consists of Cu2S + FeS (very small amount) (D) Extraction of zinc from ZnS involves roasting followed by carbon reduction.

35. Ans (A,B,C)

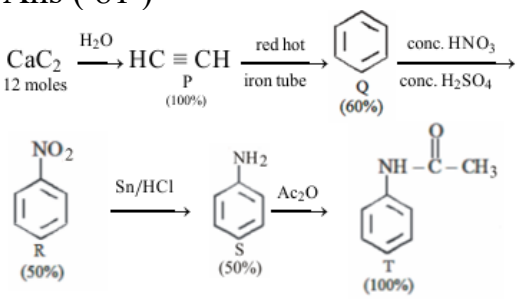
(A) Colloid-ion is a 4% solution of nitro-cellulose in a mixture of alcohol and ether. (NCERT)

(B) A mixture of milk and water appears blue when viewed by the reflected light and red when viewed by the transmitted light. (NCERT)

(C) Starch, gum, clay → Negatively charged sol.
Methylene blue → Positively charged sol.
(NCERT)

(D) Correct

36. Ans (81)



Actual yield of P = 12 moles

3 moles of P form = 1 mole of Q

So actual yield of Q = 60% of 4 = 2.4 moles

Actual yield of R = 1.2 moles

Actual yield of S = 0.6 moles

Actual yield of T = 0.6 moles

Molar mass of T = 135 g/mol

Mass of T = $135 \times 0.6 = 81$ g

37. Ans (10)

A sample can emit 10 different spectral lines when n = 5 and n = 1 maximum energy emitted is

117.5 eV

$\Delta E = 13.6z^2\left(\frac{1}{1} - \frac{1}{25}\right) = 117.5eV \Rightarrow z = 3$

Photon of minimum energy is emitted from $5 \longrightarrow 4$ transition.

$\Delta E = 13.6 \times 9 \times \left(\frac{1}{16} - \frac{1}{25}\right)$

$= \frac{13.6 \times 9 \times 9}{16 \times 25} = 2.754eV$

Work function of metal = 1.254eV

$K.E._{max.}$ of photo electrons = 1.5eV

de-Broglie wavelength of photo electron is

$\lambda = \sqrt{\frac{150}{1.5}} = \sqrt{100}A^\circ$

$\lambda = 10A^\circ$

38. Ans (84)

$H_2O \rightarrow H_2 + \frac{1}{2}O_2$ (nf = 4)

Moles of O_2 produced $= \frac{40F \times 0.75}{F \times 4} = 7.5$ moles

$C_2H_4(g) + 3O_2(g) \rightarrow 2CO_2(g) + 2H_2O(\ell)$

moles $\frac{3.5}{28} = \frac{1}{8}$ 7.5

The heat released by $\frac{1}{8}$ moles of C_2H_4 is

$= C.\Delta T$

$= 3 \times 0.5 = 1.5KJ$

$\Delta U(KJ/mol) = \frac{-1.5 \times 28}{3.5} = -12KJ/mol$

$C_2H_4(g) + 3O_2(g) \rightarrow 2CO_2(g) + 2H_2O(\ell)$

$\Delta H = \Delta U + \Delta ngRT$ (for one mole)

$= -12 + \frac{(-2) \times 8 \times 300}{1000}$

= -12-4.8 = -16.8 KJ/mol.

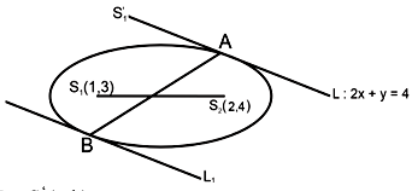
for 5 moles of ethene

$|\Delta H| = +16.8 \times 5 = 84KJ$

MATHEMATICS

39.

Ans (B)



Let $S_1'(a, b)$

$$\frac{a-1}{2} = \frac{b-3}{1} = \frac{-2(2+3-4)}{4+1}$$

$$\Rightarrow a = \frac{1}{5}, b = \frac{1}{5}$$

$$\Rightarrow S_1'\left(\frac{1}{5}, \frac{13}{5}\right)$$

Now $S_1'S_2 : y - 4 = \dfrac{\frac{13}{5} - 4}{\frac{1}{5} - 2} \quad (x - 2)$

$\therefore$ 7x - 9y = -22 ---------- (1)

Also L : 2x + y = 4 ------- (2)

By (1) and (2) : $A\left(\dfrac{14}{25}, \dfrac{72}{25}\right)$

By (1) and (2) : $A\left(\dfrac{14}{25}, \dfrac{72}{25}\right)$

Now O is mid point of A and B.

here $O\left(\dfrac{3}{2}, \dfrac{7}{2}\right)$

Let B(α, β)

$$\frac{\alpha + \frac{14}{25}}{2} = \frac{3}{2} \Rightarrow \alpha = \frac{61}{25}$$

$$\frac{\beta + \frac{72}{25}}{2} = \frac{7}{2} \Rightarrow \beta = \frac{103}{25}$$

$$\therefore B\left(\frac{61}{25}, \frac{103}{25}\right)$$

40.

Ans (B)

$$I = \int_1^2 (2^x + 2^{-x})\, d(2^x)$$

$$= \int_0^1 (2^x + 2^{-x})\, 2^x \ell n 2 dx$$

$$= \ell n 2 \int_0^1 (4^x + 1) dx$$

$$= \ell n 2 \left[\frac{4^x}{\ell n 4} + x\right]_0^1$$

$$= \ell n 2 \left[\frac{4}{\ell n 4} + 1 - \frac{1}{\ell n 4}\right]$$

$$= \ell n 2 \left[\frac{3}{\ell n 4} + 1\right]$$

$$= \frac{3}{2} + \ell n 2$$

41.

Ans (C)

Given limit

$$= \lim_{x\to 0}\left(\frac{e^{\sin x}-e^{x}}{\sin x - x}\right)\left(\frac{x}{\sin x}\right)$$

$= 1$

42.

Ans (B)

$$\sum_{i=1}^{20}(x_i-\bar{x})(y_i-\bar{y}) = 90$$

$$\Rightarrow \sum x_iy_i - \bar{x}\sum y_i - \bar{y}\sum x_i + \bar{x}\bar{y}\sum 1 = 90$$

$$\Rightarrow \sum x_iy_i - 20\bar{x}\bar{y} = 90$$

$$\Rightarrow \sigma^2 = \frac{\sum_{i=1}^{20}(x_i-y_i)^2}{20} - (\bar{x}-\bar{y})^2$$

$$= \frac{\sum x_i^2}{20} + \frac{\sum y_i^2}{20} - \frac{2\sum x_iy_i}{20} - (\bar{x})^2 - (\bar{y})^2 + 2\bar{x}\bar{y}$$

$$= 16 + 9 - \frac{2}{20}(90) = 16$$

$$\Rightarrow \sigma = 4$$

43. Ans (7.00)

$$D = \begin{vmatrix} 10 & 11 & 12 \\ 13 & 14 & 15 \\ 16 & 17 & 18 \end{vmatrix} = 0$$

$$D_1 = \begin{vmatrix} a & 11 & 12 \\ b & 14 & 15 \\ c-3 & 17 & 18 \end{vmatrix} = \begin{vmatrix} a & 11 & 1 \\ b & 14 & 1 \\ c-3 & 17 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} a & 11 & 1 \\ b-a & 3 & 0 \\ c-b-3 & 3 & 0 \end{vmatrix}$$

$$\Rightarrow 3b - 3a - 3c + 3b + 9$$

$$\Rightarrow 6b - 3a - 3c + 9$$

$$\Rightarrow 3[2b - a - c + 3]$$

$$D_2 = \begin{vmatrix} 10 & a & 12 \\ 13 & b & 15 \\ 16 & c-3 & 18 \end{vmatrix} = \begin{vmatrix} 10 & a & 12 \\ 3 & b-a & 3 \\ 3 & c-b-3 & 3 \end{vmatrix}$$

$$= \begin{vmatrix} 10 & a & 12 \\ 3 & b-a & 3 \\ 0 & c-2b+a-3 & 0 \end{vmatrix}$$

$$= -(c - 2b + a - 3)[-6] \Rightarrow 6(c + a - 2b - 3)$$

$$D_3 = \begin{vmatrix} 10 & 11 & a \\ 13 & 14 & b \\ 16 & 17 & c-3 \end{vmatrix} = \begin{vmatrix} 10 & 1 & a \\ 13 & 1 & b \\ 16 & 1 & c-3 \end{vmatrix}$$

$$= \begin{vmatrix} 10 & 1 & a \\ 3 & 0 & b-a \\ 3 & 0 & c-b-3 \end{vmatrix}$$

$$= -1[3c - 3b - 9 - 3b + 3a]$$

$$= [6b - 3c - 3a + 9]$$

$$= 3[2b + 3 - a - c]$$

system will be consistent if $D = D_1 = D_2 = D_3 = 0$

$$\Rightarrow 2b + 3 - a - c = 0$$

Now $|X| = a(-1) - 2(-b) + 1(-c)$

$$= -a + 2b - c$$

$$= -3$$

Now plane $P : -x + 2y - z + 3 = 0$

44. Ans (24.00)

$$D = \begin{vmatrix} 10 & 11 & 12 \\ 13 & 14 & 15 \\ 16 & 17 & 18 \end{vmatrix} = 0$$

$$D_1 = \begin{vmatrix} a & 11 & 12 \\ b & 14 & 15 \\ c-3 & 17 & 18 \end{vmatrix} = \begin{vmatrix} a & 11 & 1 \\ b & 14 & 1 \\ c-3 & 17 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} a & 11 & 1 \\ b-a & 3 & 0 \\ c-b-3 & 3 & 0 \end{vmatrix}$$

$\Rightarrow 3b - 3a - 3c + 3b + 9$

$\Rightarrow 6b - 3a - 3c + 9$

$\Rightarrow 3[2b - a - c + 3]$

$$D_2 = \begin{vmatrix} 10 & a & 12 \\ 13 & b & 15 \\ 16 & c-3 & 18 \end{vmatrix} = \begin{vmatrix} 10 & a & 12 \\ 3 & b-a & 3 \\ 3 & c-b-3 & 3 \end{vmatrix}$$

$$= \begin{vmatrix} 10 & a & 12 \\ 3 & b-a & 3 \\ 0 & c-2b+a-3 & 0 \end{vmatrix}$$

$= -(c - 2b + a - 3)[-6] \Rightarrow 6(c + a - 2b - 3)$

$$D_3 = \begin{vmatrix} 10 & 11 & a \\ 13 & 14 & b \\ 16 & 17 & c-3 \end{vmatrix} = \begin{vmatrix} 10 & 1 & a \\ 13 & 1 & b \\ 16 & 1 & c-3 \end{vmatrix}$$

$= -1[3c - 3b - 9 - 3b + 3a]$

$= [6b - 3c - 3a + 9]$

$= 3[2b + 3 - a - c]$

system will be consistent if $D = D_1 = D_2 = D_3 = 0$

$\Rightarrow 2b + 3 - a - c = 0$

Now $|X| = a(-1) - 2(-b) + 1(-c)$

$= -a + 2b - c$

$= -3$

Now plane $P : -x + 2y - z + 3 = 0$

45. Ans (26.00)

$f(x)f(y) + 2 = f(x) + f(y) + f(xy)$ ....(1)

putting $x = y = 1$

$f^2(1) + 2 = 3f(1)$

$f^2(1) - 3f(1) + 2 = 0$

$f(1) = 1, 2$

since $f(x)$ is one-one & $f(0) = 1$

so $f(1) = 2$

replace y with $\frac{1}{x}$ in (1)

$$f(x)f\left(\frac{1}{x}\right) + 2 = f(x) + f\left(\frac{1}{x}\right) + 2$$

$f(x) = 1 \pm x^n$

but $f(1) = 2$ & $f'(1) = 2$

$\therefore f(x) = 1 + x^2$

46. Ans (3.00)

$f(x)f(y) + 2 = f(x) + f(y) + f(xy)$ ....(1)

putting $x = y = 1$

$f^2(1) + 2 = 3f(1)$

$f^2(1) - 3f(1) + 2 = 0$

$f(1) = 1, 2$

since $f(x)$ is one-one & $f(0) = 1$

so $f(1) = 2$

replace y with $\frac{1}{x}$ in (1)

$f(x)f\left(\frac{1}{x}\right) + 2 = f(x) + f\left(\frac{1}{x}\right) + 2$

$f(x) = 1 \pm x^n$

but $f(1) = 2$ & $f'(1) = 2$

$\therefore f(x) = 1 + x^2$

47. Ans (3.00)

When $n = 1$, $a_1 = 5S_1 + 1 = 5a_1 + 1$

$\Rightarrow a_1 = \frac{-1}{4}$

for $n \geqslant 1$

$a_{n+1} - a_n = 5S_{n+1} - 5S_n$

$\Rightarrow a_{n+1} - a_n = 5a_{n+1}$

$\Rightarrow 4a_{n+1} = -a_n \Rightarrow \frac{a_{n+1}}{a_n} = \frac{-1}{4}$

there fore $\{a_n\}$ is a GP with $a_1 = \frac{-1}{4}$ & common ratio $r = \frac{-1}{4}$

So $a_{10} = \left(\frac{-1}{4}\right)\left(\frac{-1}{4}\right)^9 = \frac{1}{4^{10}}$

Also $a_{12} = \left(\frac{-1}{4}\right)\left(\frac{-1}{4}\right)^{11} = \frac{1}{4^{12}}$

So $b_{12} = \frac{4 + a_{12}}{1 - a_{12}} \Rightarrow \frac{4 + \frac{1}{4^{12}}}{1 - \frac{1}{4^{12}}} \Rightarrow \frac{4^{13} + 1}{4^{12} - 1}$

48.

When n = 1, $a_1 = 5S_1 + 1 = 5a_1 + 1$

$\Rightarrow a_1 = \frac{-1}{4}$

for $n \geqslant 1$

$a_{n+1} - a_n = 5\,S_{n+1} - 5S_n$

$\Rightarrow a_{n+1} - a_n = 5a_{n+1}$

$\Rightarrow 4a_{n+1} = -a_n \Rightarrow \frac{a_{n+1}}{a_n} = \frac{-1}{4}$

there fore $\{a_n\}$ is a GP with $a_1 = \frac{-1}{4}$ & common ratio $r = \frac{-1}{4}$

So $a_{10} = \left(\frac{-1}{4}\right)\left(\frac{-1}{4}\right)^9 = \frac{1}{4^{10}}$

Also $a_{12} = \left(\frac{-1}{4}\right)\left(\frac{-1}{4}\right)^{11} = \frac{1}{4^{12}}$

So $b_{12} = \frac{4 + a_{12}}{1 - a_{12}} \Rightarrow \frac{4 + \frac{1}{4^{12}}}{1 - \frac{1}{4^{12}}} \Rightarrow \frac{4^{13} + 1}{4^{12} - 1}$

49. Ans (A,C)

$$I = \int \frac{3x^4 - 2(x^4 - 16)}{x^8(x^4 - 16)}dx$$

$$= 3\int \frac{1}{x^4(x^4 - 16)}dx - 2\int \frac{1}{x^8}dx$$

$$= \frac{3}{16}\int \frac{x^4 - (x^4 - 16)}{x^4(x^4 - 16)}dx + \frac{2}{7}\cdot\frac{1}{x^7}$$

$$= \frac{3}{16}\int \frac{1}{x^4 - 16}dx - \frac{3}{16}\int \frac{1}{x^4}dx + \frac{2}{7x^7}$$

$$= \frac{1}{8} \times \frac{3}{16}\int \frac{(x^2 + 4) - (x^2 - 4)}{(x^2 + 4)(x^2 - 4)}dx + \frac{3}{16}\cdot\frac{1}{3x^3} + \frac{2}{7x^7}$$

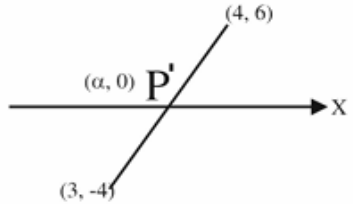
$$= \frac{3}{128}\int \left(\frac{1}{x^2 - 4} - \frac{1}{x^2 + 4}\right)dx + \frac{1}{16x^3} + \frac{2}{7x^7}$$

$$= \frac{3}{128}\left(\frac{1}{2.2}\ell n\left|\frac{x-2}{x+2}\right| - \frac{1}{2}\tan^{-1}\frac{x}{2}\right) + \frac{1}{16x^3} + \frac{2}{7x^7} + C$$

$$= \frac{3}{512}\ell n\left|\frac{x-2}{x+2}\right| - \frac{3}{256}\tan^{-1}\frac{x}{2} + \frac{1}{16x^3} + \frac{2}{7x^7} + C$$

51. Ans (B,C,D)

$PA + PB = \sqrt{(\alpha-3)^2+16} + \sqrt{(\alpha-4)^2+36}$



which is equivalent to P'A' + P'B' where P'(α, 0),

A'(3, –4) & B'(4, 6)

it will be minimum when

$\frac{4}{\alpha-3} = \frac{10}{1} \Rightarrow 2 = 5\alpha - 15$

$\Rightarrow \alpha = \frac{17}{5}$

So P $\left(\frac{17}{5}, 0, 0\right)$

let Q (0, λ, 0)

Now $QA + QB = \sqrt{\lambda^2+25} + \sqrt{(\lambda-6)^2+16}$

it will be minimum when $\lambda = \frac{10}{3}$

$\therefore$ Q $\left(0, \frac{10}{3}, 0\right)$